

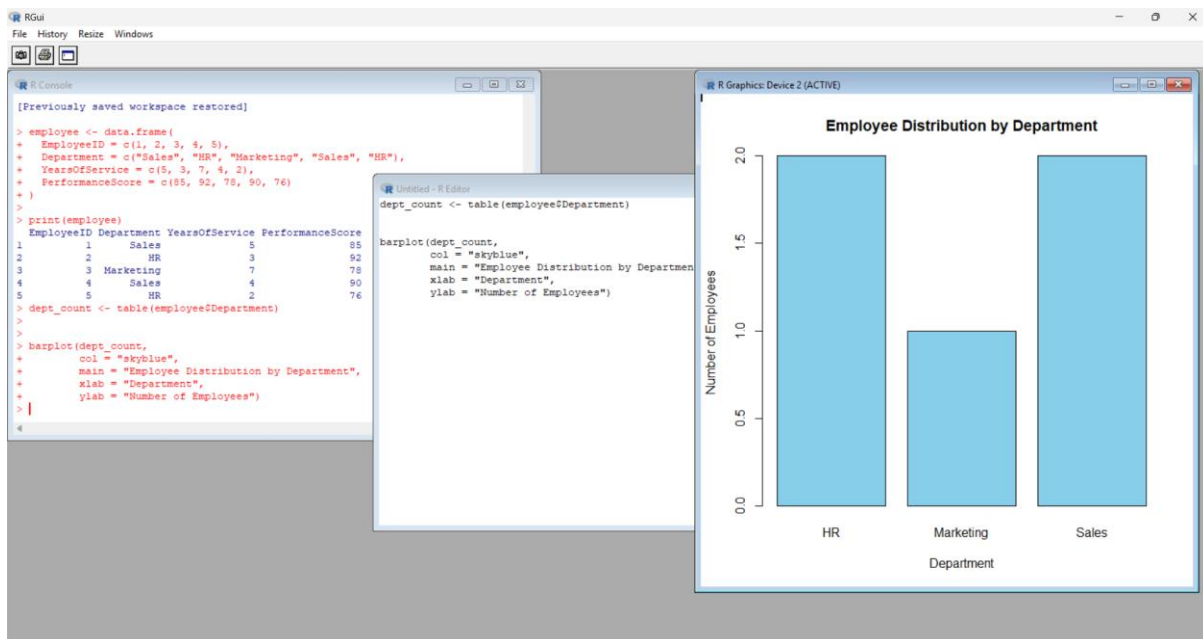
23-30

Set 23: Employee Performance Analysis

23.1. In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.

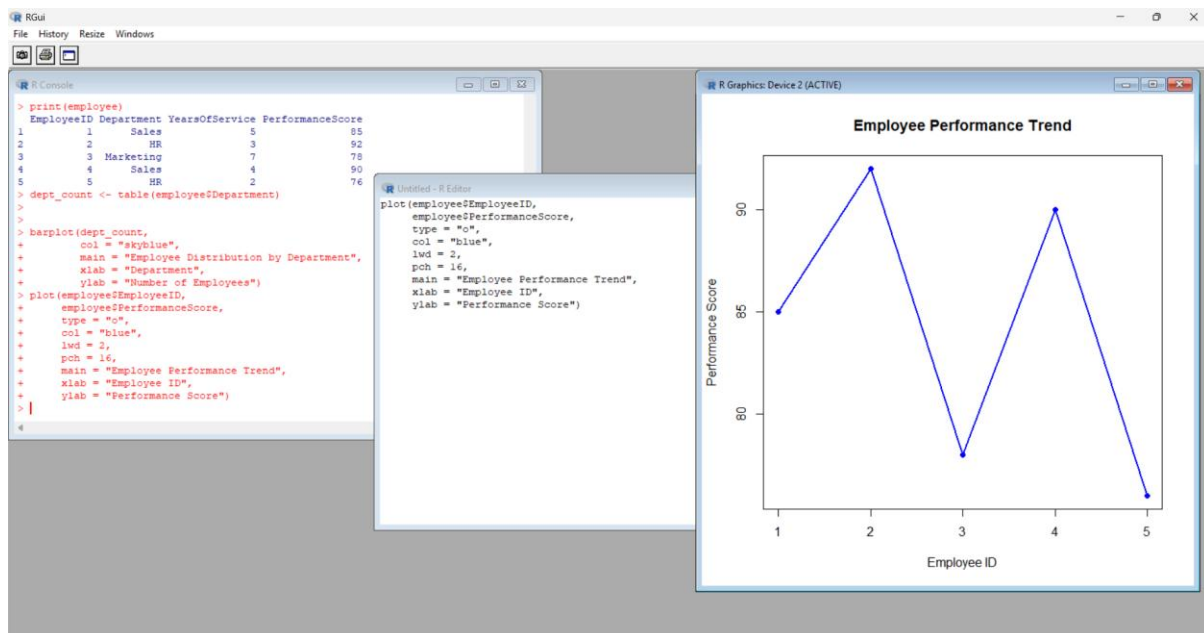
```
dept_count <- table(employee$Department)
```

```
barplot(dept_count,  
        col = "skyblue",  
        main = "Employee Distribution by Department",  
        xlab = "Department",  
        ylab = "Number of Employees")
```



23.2. In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

```
plot(employee$EmployeeID,  
employee$PerformanceScore,  
      type = "o",  
      col = "blue",  
      lwd = 2,  
      pch = 16,  
      main = "Employee Performance Trend",  
      xlab = "Employee ID",  
      ylab = "Performance Score")
```



23.3. In Tableau, build a dashboard combining a bar chart showing department distribution and a scatter plot displaying the relationship between years of service and performance scores.

```
par(mfrow = c(1,2))
```

```
barplot(dept_count,
```

```
col = "lightgreen",
```

```
main = "Department Distribution",
```

```
xlab = "Department",
```

```
ylab = "Count")
```

```
plot(employee$YearsOfService,
```

```
employee$PerformanceScore,
```

```
pch = 19,
```

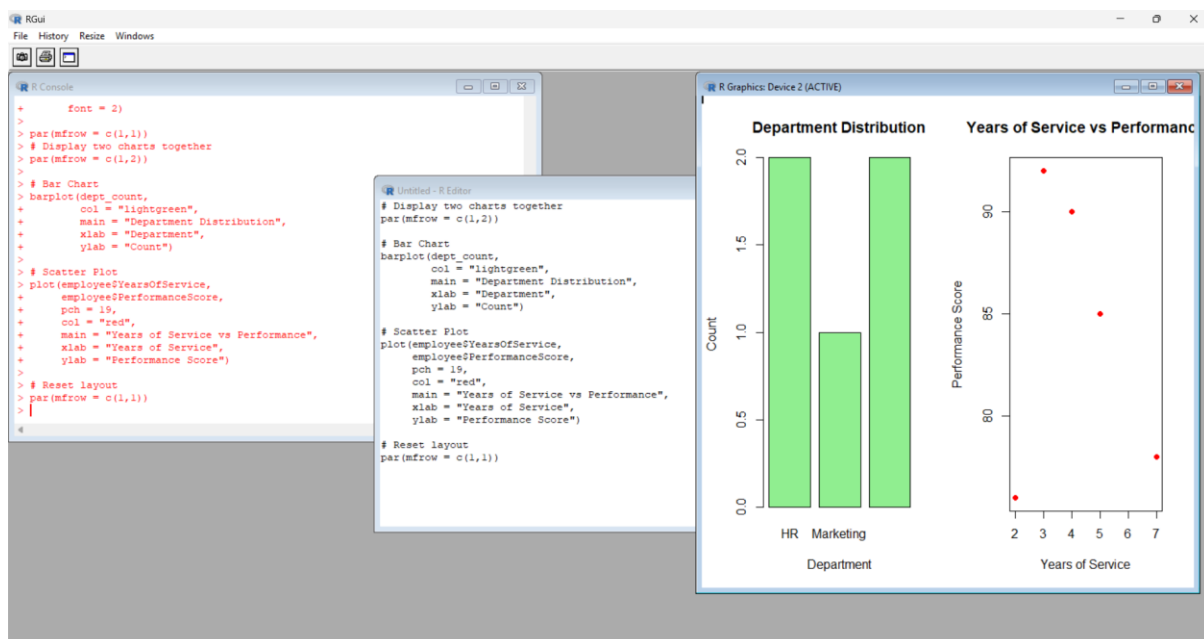
```
col = "red",
```

```
main = "Years of Service vs Performance",
```

```
xlab = "Years of Service",
```

```
ylab = "Performance Score")
```

```
par(mfrow = c(1,1))
```



23.4. In R, develop a table showing the performance data for each employee. Label the table elements.

```
employee <- data.frame(  
  EmployeeID = c(1, 2, 3, 4, 5),  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  YearsOfService = c(5, 3, 7, 4, 2),  
  PerformanceScore = c(85, 92, 78, 90, 76)  
)
```

View(employee)

print(employee)

The screenshot displays the R GUI environment. The R Console window shows the following code being executed:

```
> pch = 19,  
> col = "red",  
> main = "Years of Service vs Performance",  
> xlab = "Years of Service",  
> ylab = "Performance Score")  
>  
> # Reset layout  
> par(mfrow = c(1,1))  
> employee <- data.frame(  
+ EmployeeID = c(1, 2, 3, 4, 5),  
+ Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
+ YearsOfService = c(5, 3, 7, 4, 2),  
+ PerformanceScore = c(85, 92, 78, 90, 76)  
)  
>  
> View(employee)  
>  
> print(employee)  
EmployeeID Department YearsOfService PerformanceScore  
1 1 Sales 5 85  
2 2 HR 3 92  
3 3 Marketing 7 78  
4 4 Sales 4 90  
5 5 HR 2 76  
> |  
4
```

The R Editor window shows the code for creating the data frame and the View/print commands:

```
employee <- data.frame(  
  EmployeeID = c(1, 2, 3, 4, 5),  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  YearsOfService = c(5, 3, 7, 4, 2),  
  PerformanceScore = c(85, 92, 78, 90, 76)  
)  
  
View(employee)  
  
print(employee)
```

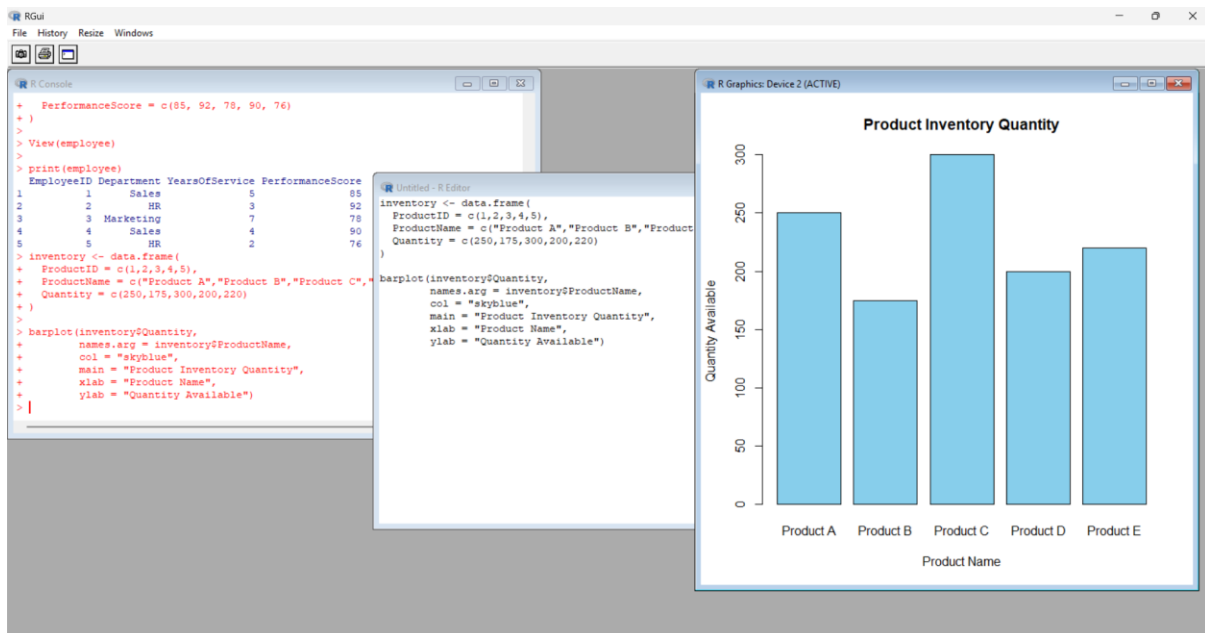
A 'Data: employee' window displays the resulting data frame as a table:

	EmployeeID	Department	YearsOfService	PerformanceScore
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78
4	4	Sales	4	90
5	5	HR	2	76

Set 24 Product Inventory Management

24.1. In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
inventory <- data.frame(  
  ProductID = c(1,2,3,4,5),  
  ProductName = c("Product A","ProductB","ProductC","ProductD","Product E"),  
  Quantity = c(250,175,300,200,220)  
)  
  
barplot(inventory$Quantity,  
  names.arg = inventory$ProductName,  
  col = "skyblue",  
  main = "Product Inventory Quantity",  
  xlab = "Product Name",  
  ylab = "Quantity Available")
```



24.2. In R, generate a stacked bar chart to show the quantity of each product within different product categories.

```
qty <- matrix(inventory$Quantity, nrow = 1)
```

```
colnames(qty) <- inventory$ProductName
```

```
rownames(qty) <- "Quantity"
```

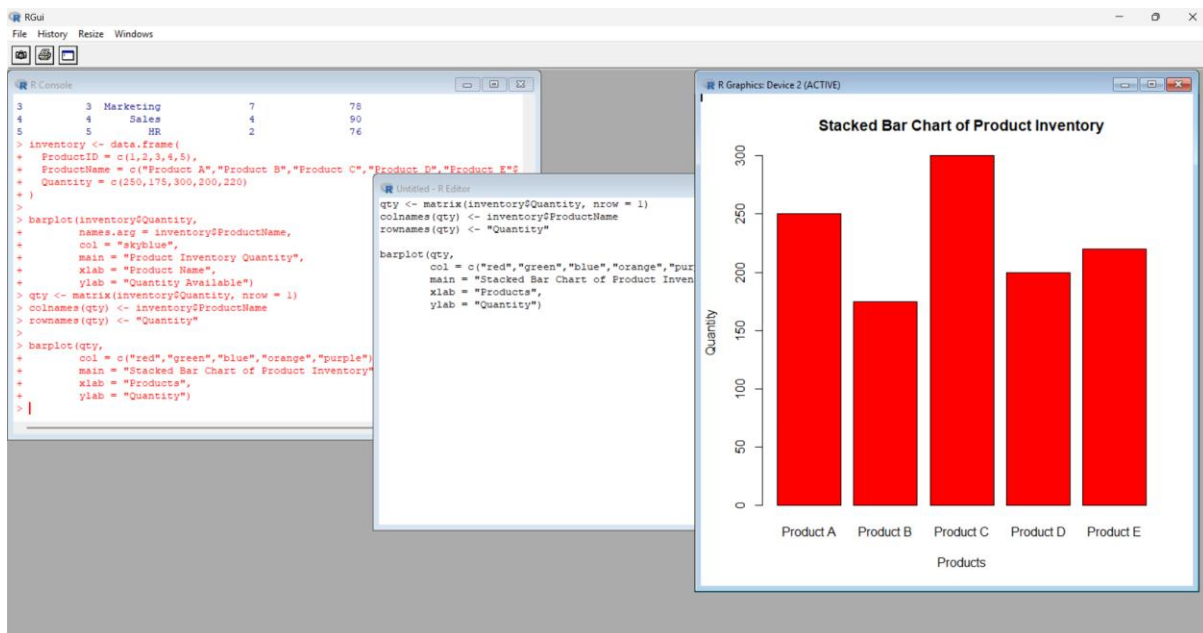
```
barplot(qty,
```

```
col = c("red", "green", "blue", "orange", "purple"),
```

```
main = "Stacked Bar Chart of Product Inventory",
```

```
xlab = "Products",
```

```
ylab = "Quantity")
```



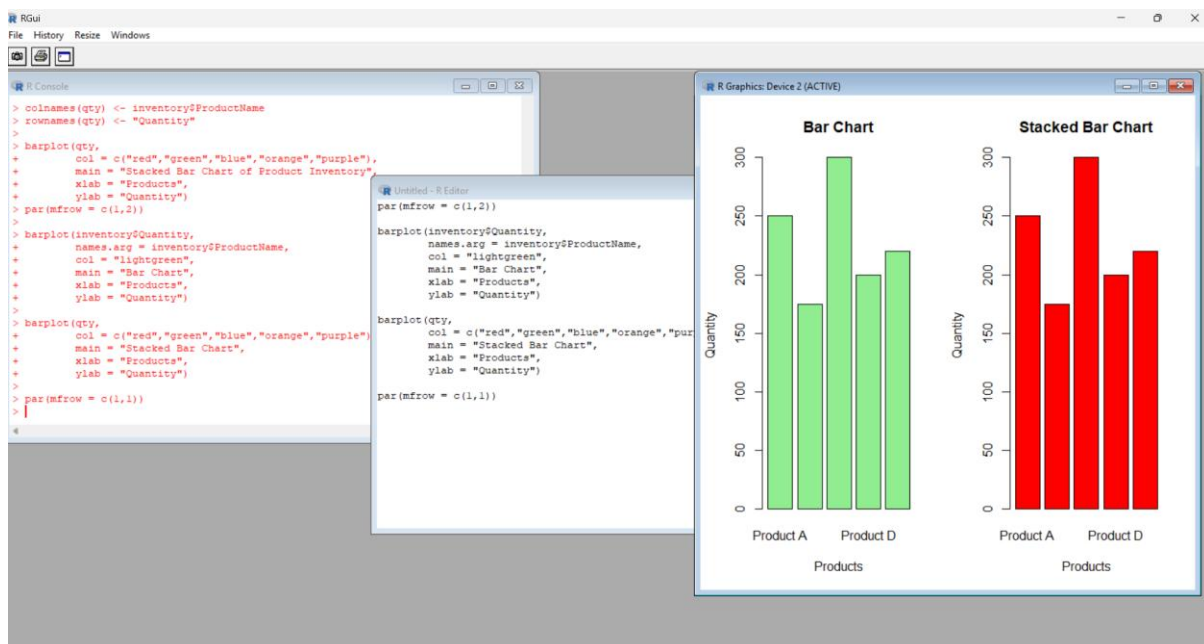
24.3. In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive exploration of inventory data.

```
par(mfrow = c(1,2))
```

```
barplot(inventory$Quantity,  
names.arg = inventory$ProductName,  
col = "lightgreen",  
main = "Bar Chart",  
xlab = "Products",  
ylab = "Quantity")
```

```
barplot(qty,  
col = c("red","green","blue","orange","purple"),  
main = "Stacked Bar Chart",  
xlab = "Products",  
ylab = "Quantity")
```

```
par(mfrow = c(1,1))
```



24.4. In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.

```
inventory$Price<- c(120, 150, 180, 140, 160)
```

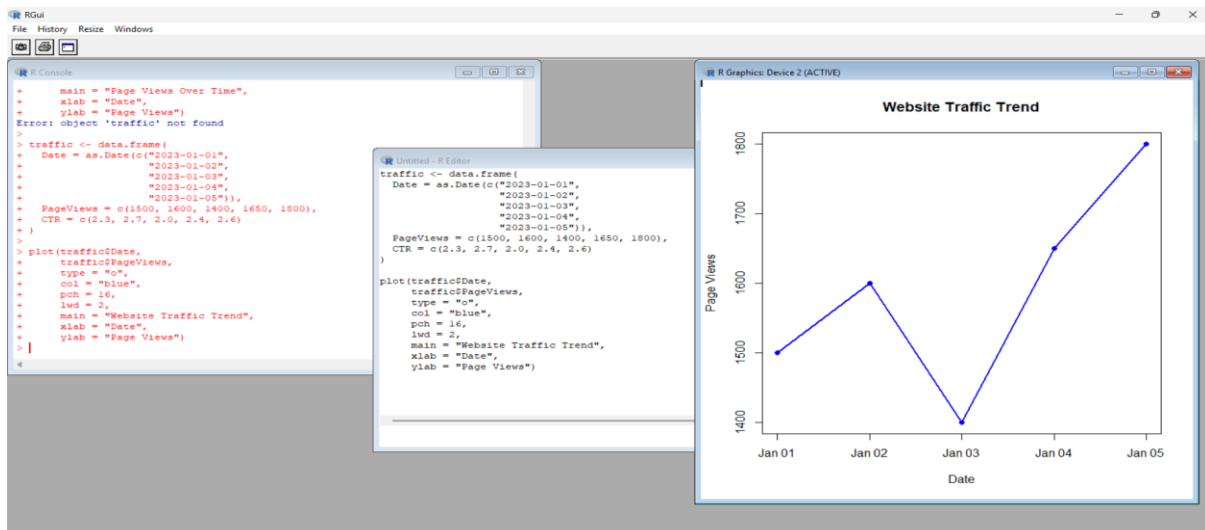
```
plot(inventory$Price,  
inventory$Quantity,  
pch = 19,  
col = "red",  
main = "Product Price vs Quantity Available",  
xlab = "Product Price",  
ylab = "Quantity Available")
```



Set 25: Website Traffic Analysis

25.1. In Tableau, create a line chart to visualize the trend in page views over time. Label the axes and title the chart.

```
traffic <- data.frame(  
  Date = as.Date(c("2023-01-01",  
    "2023-01-02",  
    "2023-01-03",  
    "2023-01-04",  
    "2023-01-05")),  
  PageViews = c(1500, 1600, 1400, 1650, 1800),  
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)  
)  
plot(traffic$Date,  
  traffic$PageViews,  
  type = "o",  
  col = "blue",  
  pch = 16,  
  lwd = 2,  
  main = "Website Traffic Trend",  
  xlab = "Date",  
  ylab = "Page Views")
```



25.2. In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.

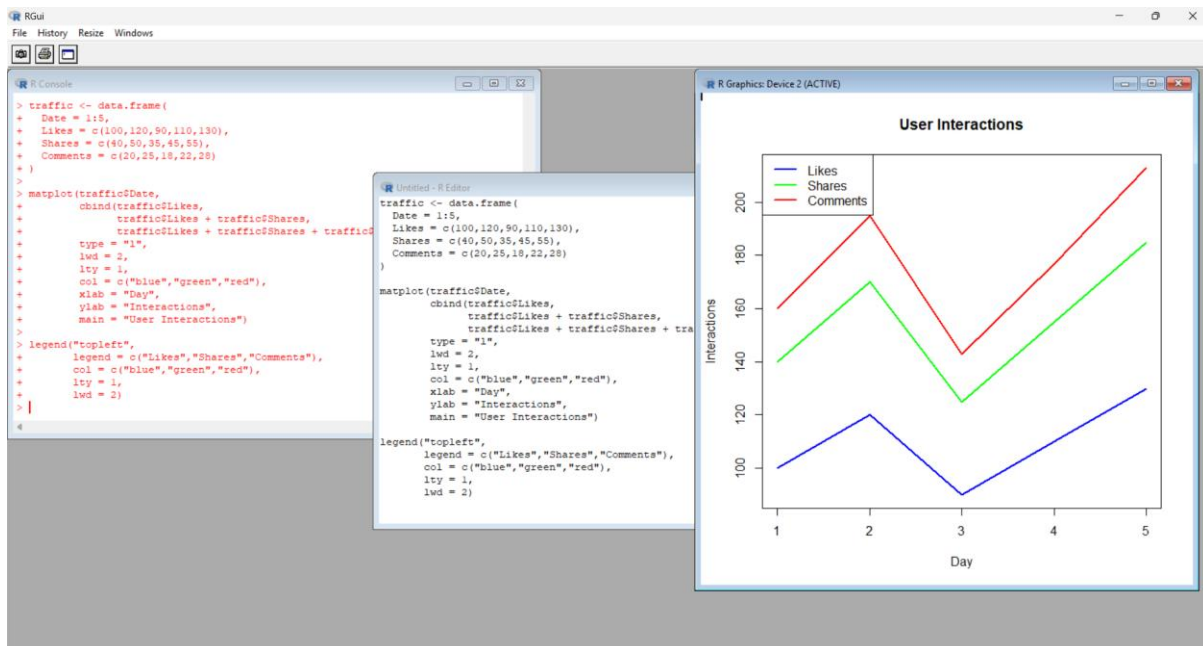
```
traffic <- data.frame(  
  Date = c("2023-01-01", "2023-01-02", "2023-01-03",  
           "2023-01-04", "2023-01-05"),  
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)  
)
```

```
barplot(traffic$CTR,  
names.arg = traffic$Date,  
  col = "orange",  
  main = "Click-through Rate by Date",  
xlab = "Date",  
ylab = "CTR (%)")
```



25.3. In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.

```
traffic <- data.frame(  
  Date = 1:5,  
  Likes = c(100,120,90,110,130),  
  Shares = c(40,50,35,45,55),  
  Comments = c(20,25,18,22,28)  
)  
  
matplot(traffic$Date,  
  cbind(traffic$Likes,  
  traffic$Likes + traffic$Shares,  
  traffic$Likes + traffic$Shares + traffic$Comments),  
  type = "l",  
  lwd = 2,  
  lty = 1,  
  col = c("blue","green","red"),  
  xlab = "Day",  
  ylab = "Interactions",  
  main = "User Interactions")  
  
legend("topleft",  
  legend = c("Likes","Shares","Comments"),  
  col = c("blue","green","red"),  
  lty = 1,  
  lwd = 2)
```



25.4. In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source.

```
traffic <- data.frame(
  Date = as.Date(c("2023-01-01",
    "2023-01-02",
    "2023-01-03",
    "2023-01-04",
    "2023-01-05")),
  PageViews = c(1500,1600,1400,1650,1800),
  CTR = c(2.3,2.7,2.0,2.4,2.6)
)

par(mfrow = c(1,2))

plot(traffic$Date,
traffic$PageViews,
  type = "o",
```

```

    col = "blue",

pch = 16,

lwd = 2,

    main = "Page Views",

xlab = "Date",

ylab = "Page Views")

barplot(traffic$CTR,

names.arg = traffic$Date,

    col = "orange",

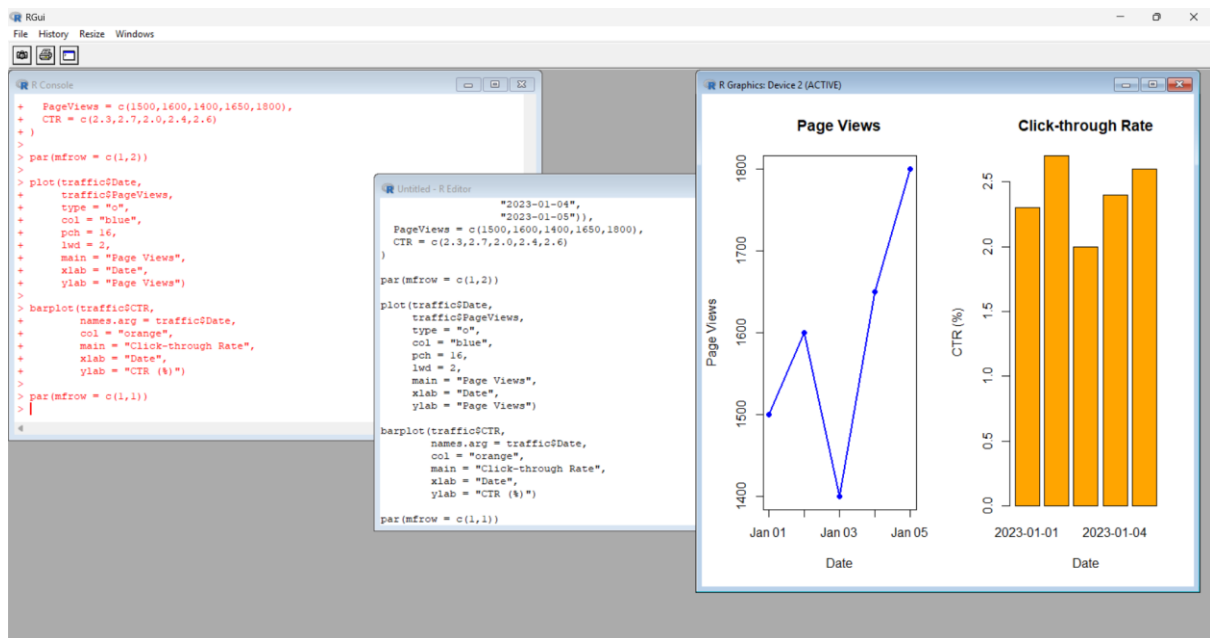
    main = "Click-through Rate",

xlab = "Date",

ylab = "CTR (%)")

par(mfrow = c(1,1))

```



SET 26

26.1. Import the dataset in R. Plot a histogram for Math Score. Draw a boxplot of Science Score by Gender. Add proper title and labels. Interpret the results.

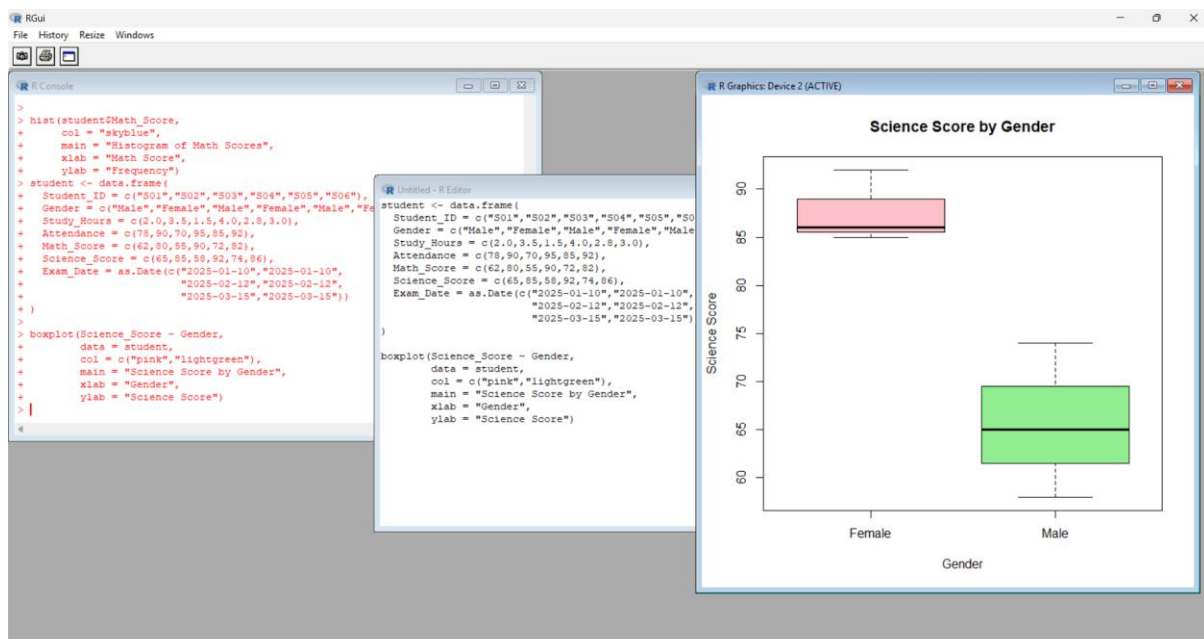
```
student <- data.frame(  
  Student_ID = c("S01","S02","S03","S04","S05","S06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Study_Hours = c(2.0,3.5,1.5,4.0,2.8,3.0),  
  Attendance = c(78,90,70,95,85,92),  
  Math_Score = c(62,80,55,90,72,82),  
  Science_Score = c(65,85,58,92,74,86),  
  Exam_Date = as.Date(c("2025-01-10","2025-01-10",  
    "2025-02-12","2025-02-12",  
    "2025-03-15","2025-03-15"))  
)  
  
hist(student$Math_Score,  
  col = "skyblue",  
  main = "Histogram of Math Scores",  
  xlab = "Math Score",  
  ylab = "Frequency")
```



Box Plot:

```
student <- data.frame(  
  Student_ID = c("S01", "S02", "S03", "S04", "S05", "S06"),  
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),  
  Study_Hours = c(2.0, 3.5, 1.5, 4.0, 2.8, 3.0),  
  Attendance = c(78, 90, 70, 95, 85, 92),  
  Math_Score = c(62, 80, 55, 90, 72, 82),  
  Science_Score = c(65, 85, 58, 92, 74, 86),  
  Exam_Date = as.Date(c("2025-01-10", "2025-01-10",  
    "2025-02-12", "2025-02-12",  
    "2025-03-15", "2025-03-15"))  
)
```

```
boxplot(Science_Score ~ Gender,  
  data = student,  
  col = c("pink", "lightgreen"),  
  main = "Science Score by Gender",  
  xlab = "Gender",  
  ylab = "Science Score")
```



26.2. Create a scatter plot using R programming between Study Hours and Math Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.

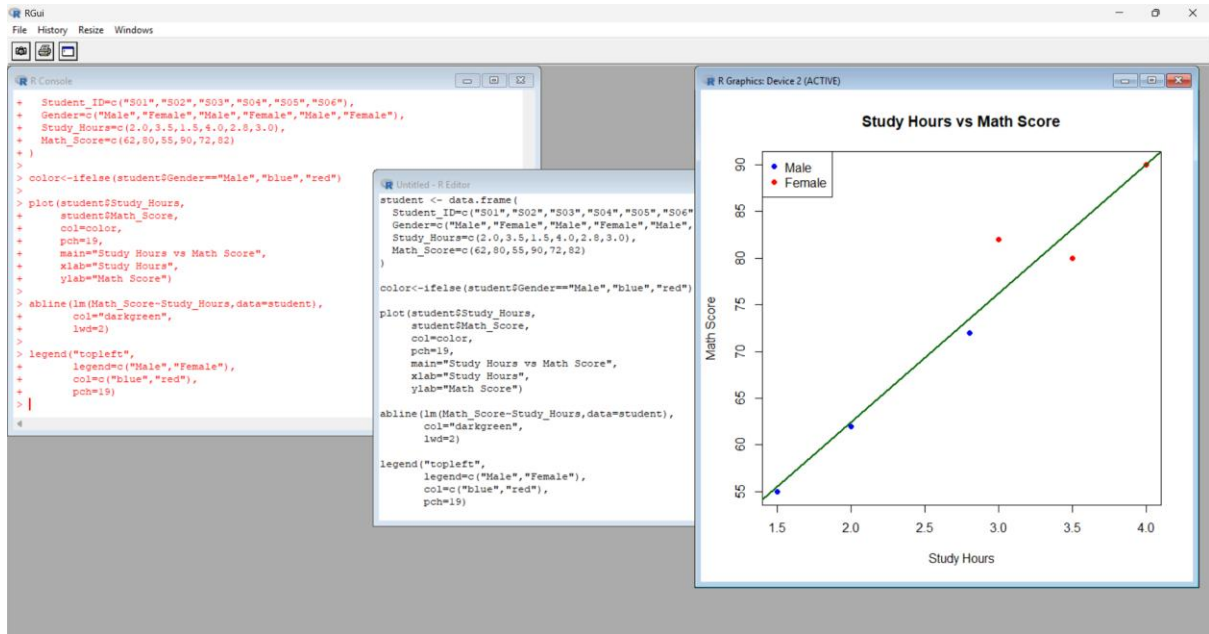
```
student <- data.frame(  
  Student_ID=c("S01","S02","S03","S04","S05","S06"),  
  Gender=c("Male","Female","Male","Female","Male","Female"),  
  Study_Hours=c(2.0,3.5,1.5,4.0,2.8,3.0),  
  Math_Score=c(62,80,55,90,72,82)  
)
```

```
color<-ifelse(student$Gender=="Male","blue","red")
```

```
plot(student$Study_Hours,  
  student$Math_Score,  
  col=color,  
  pch=19,  
  main="Study Hours vs Math Score",  
  xlab="Study Hours",  
  ylab="Math Score")
```

```
abline(lm(Math_Score~Study_Hours,data=student),  
  col="darkgreen",  
  lwd=2)
```

```
legend("topleft",  
  legend=c("Male","Female"),  
  col=c("blue","red"),  
  pch=19)
```

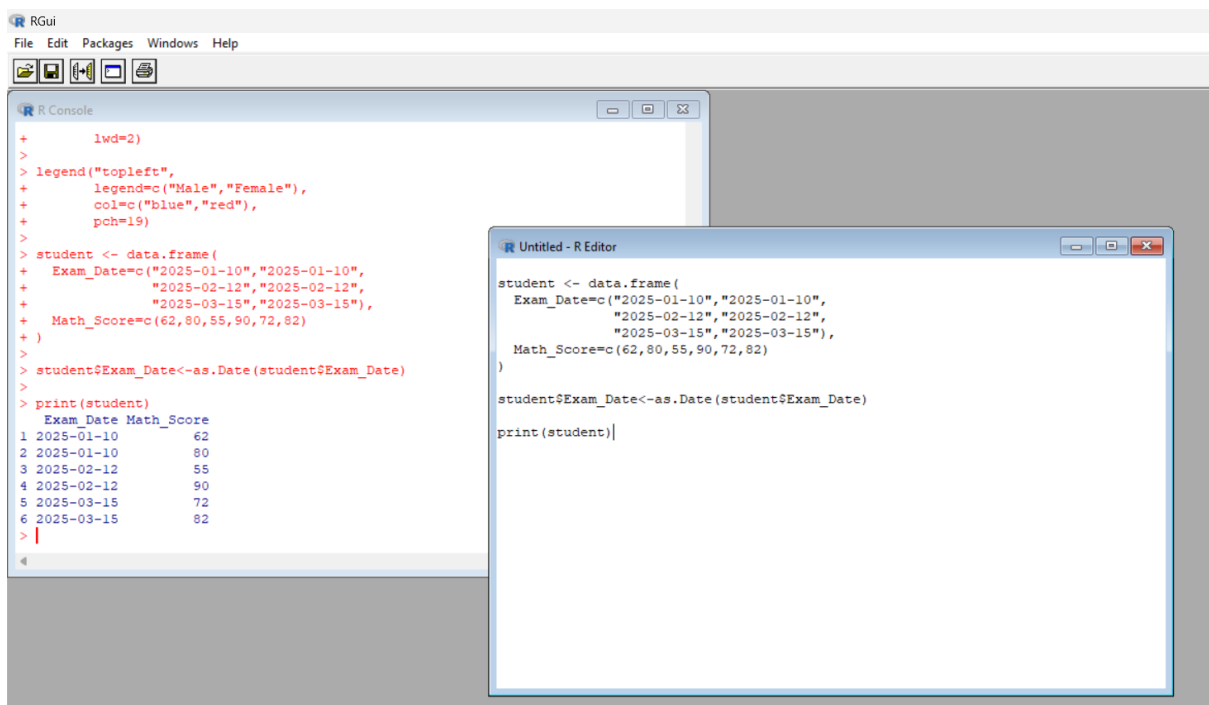



26.3. Using R, Convert Exam Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.

```
student <- data.frame(
  Exam_Date=c("2025-01-10","2025-01-10",
              "2025-02-12","2025-02-12",
              "2025-03-15","2025-03-15"),
  Math_Score=c(62,80,55,90,72,82)
)

student$Exam_Date<-as.Date(student$Exam_Date)

print(student)
```



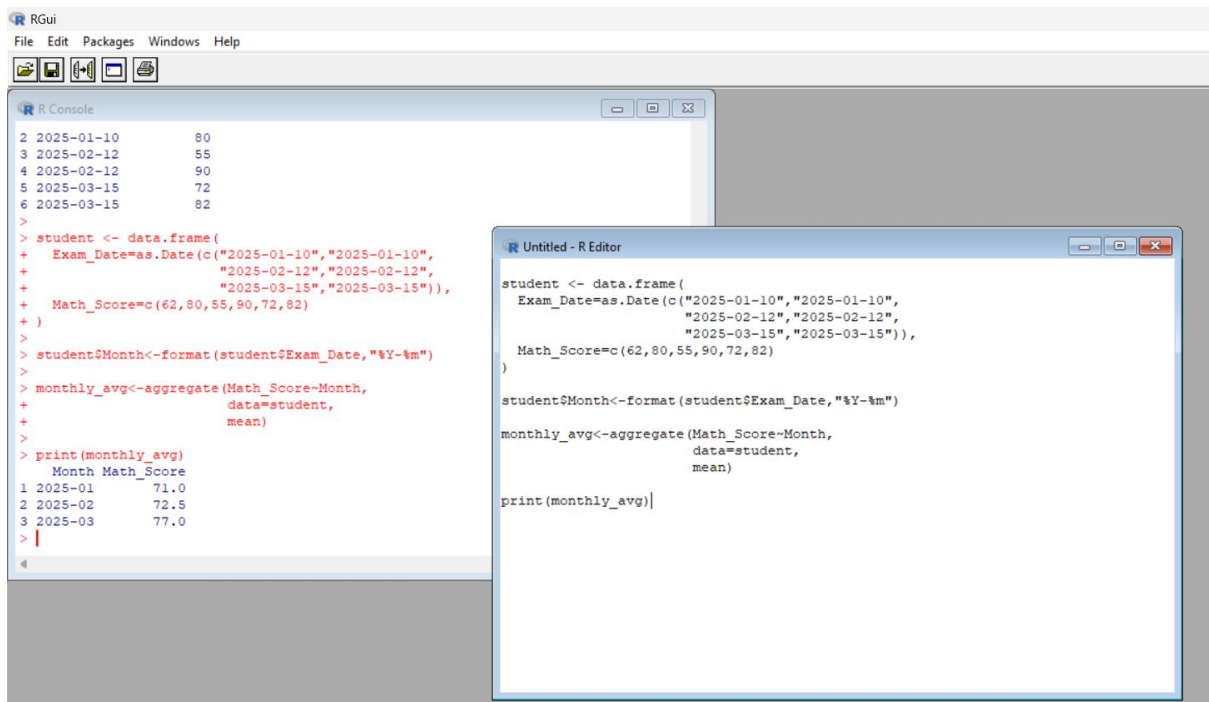
Calculate Monthly Average Math Score

```
student <- data.frame(
  Exam_Date=as.Date(c("2025-01-10","2025-01-10",
                      "2025-02-12","2025-02-12",
                      "2025-03-15","2025-03-15")),
  Math_Score=c(62,80,55,90,72,82)
)

student$Month<-format(student$Exam_Date,"%Y-%m")

monthly_avg<-aggregate(Math_Score~Month,
                        data=student,
                        mean)

print(monthly_avg)
```



Line Chart

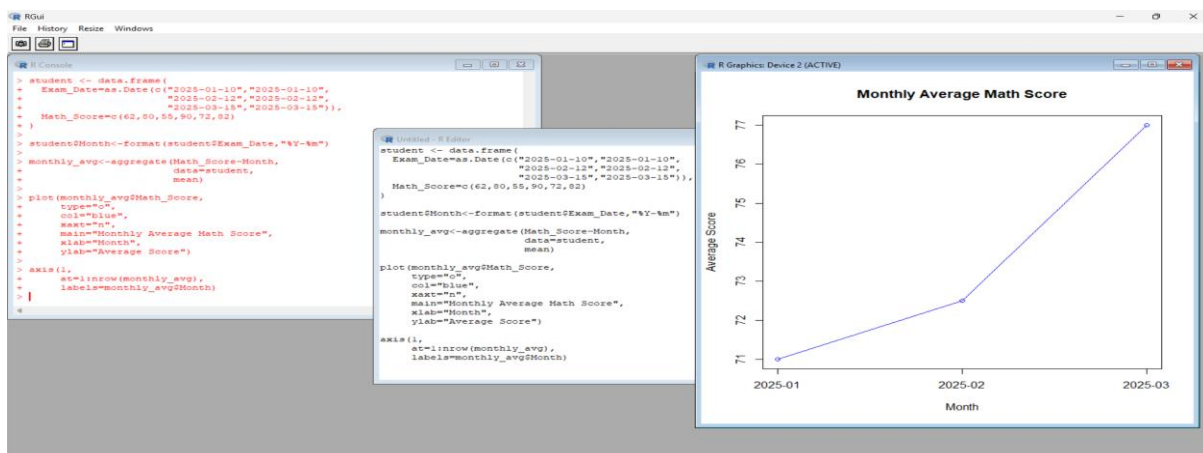
```
student <- data.frame(
  Exam_Date=as.Date(c("2025-01-10","2025-01-10",
                      "2025-02-12","2025-02-12",
                      "2025-03-15","2025-03-15")),
  Math_Score=c(62,80,55,90,72,82)
)

student$Month<-format(student$Exam_Date,"%Y-%m")

monthly_avg<-aggregate(Math_Score~Month,
                        data=student,
                        mean)

plot(monthly_avg$Math_Score,
     type="o",
     col="blue",
     xaxt="n",
     main="Monthly Average Math Score",
     xlab="Month",
     ylab="Average Score")

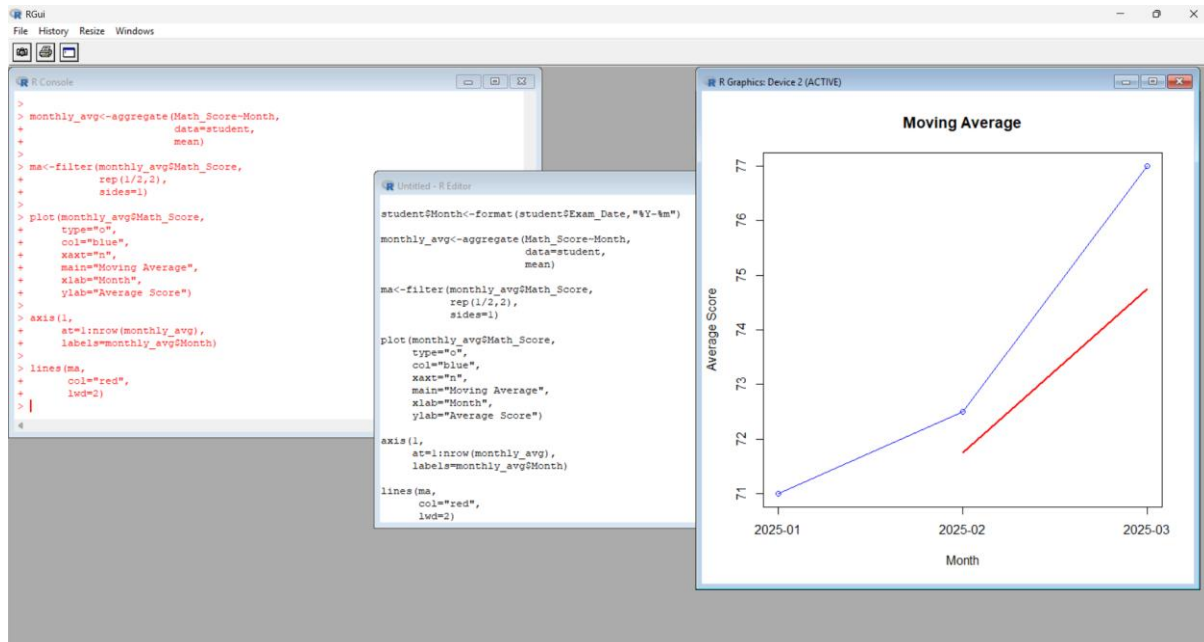
axis(1,
     at=1:nrow(monthly_avg),
     labels=monthly_avg$Month)
```



Moving Average Smoothing

```
student <- data.frame(  
  Exam_Date=as.Date(c("2025-01-10","2025-01-10",  
    "2025-02-12","2025-02-12",  
    "2025-03-15","2025-03-15")),  
  Math_Score=c(62,80,55,90,72,82)  
)  
  
student$Month<-format(student$Exam_Date,"%Y-%m")  
  
monthly_avg<-aggregate(Math_Score~Month,  
  data=student,  
  mean)  
  
ma<-filter(monthly_avg$Math_Score,  
  rep(1/2,2),  
  sides=1)  
  
plot(monthly_avg$Math_Score,  
  type="o",  
  col="blue",  
  xaxt="n",  
  main="Moving Average",  
  xlab="Month",  
  ylab="Average Score")  
  
axis(1,  
  at=1:nrow(monthly_avg),  
  labels=monthly_avg$Month)
```

```
lines(ma,
      col="red",
      lwd=2)
```



26.4. Import Student_Mini_Data.csv into Tableau. Create a bar chart showing average Math scores by Gender. Create a line chart for score trends over time. Add a filter for Exam Date. Design asimple interactive dashboard.

```
student <- data.frame(
  Student_ID = c("S01","S02","S03","S04","S05","S06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Math_Score = c(62,80,55,90,72,82),
  Exam_Date = as.Date(c("2025-01-10","2025-01-10",
                        "2025-02-12","2025-02-12",
                        "2025-03-15","2025-03-15"))
)
```

```
avg<- aggregate(Math_Score ~ Gender, data = student, mean)
```

```
par(mfrow = c(1,2))
```

```
barplot(avg$Math_Score,
```

```
names.arg = avg$Gender,
```

```
col = c("skyblue","pink"),
```

```
main = "Average Math Score by Gender",
```

```
xlab = "Gender",
```

```
ylab = "Average Math Score")
```

```
plot(student$Exam_Date,
```

```
student$Math_Score,
```

```
type = "o",
```

```
pch = 16,
```

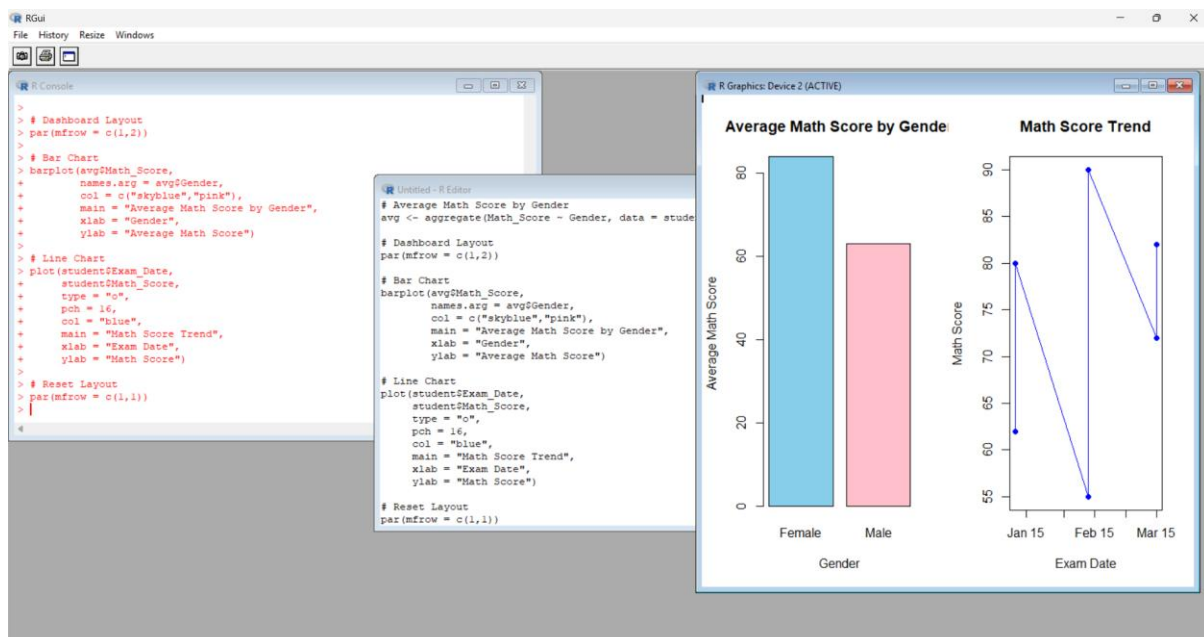
```
col = "blue",
```

```
main = "Math Score Trend",
```

```
xlab = "Exam Date",
```

```
ylab = "Math Score")
```

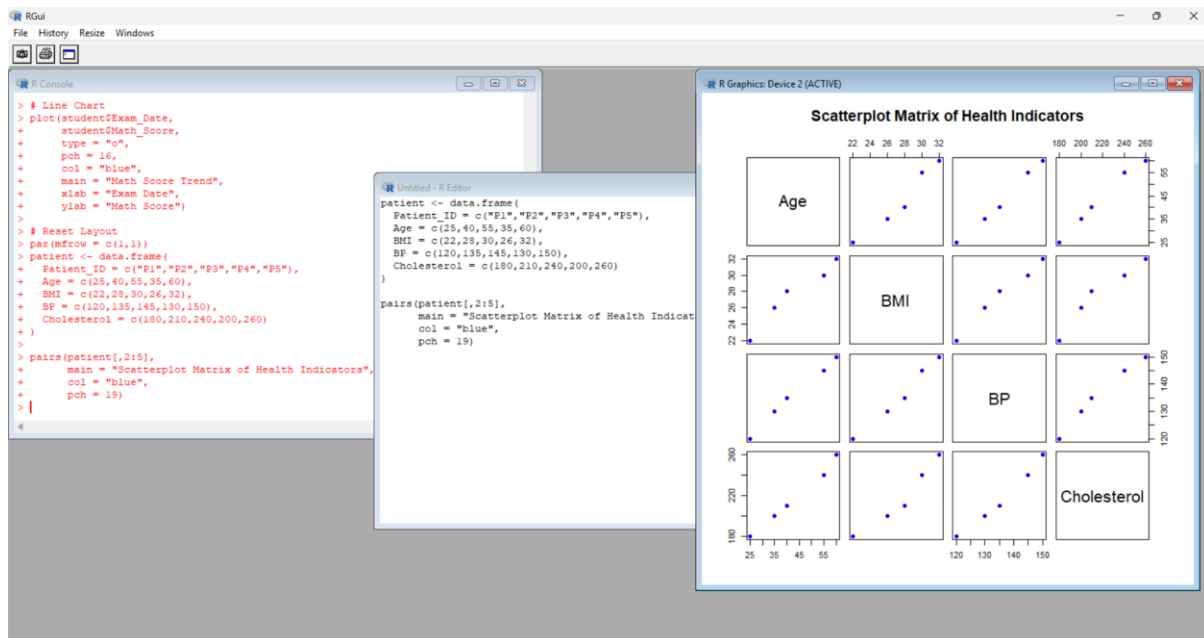
```
par(mfrow = c(1,1))
```



SET 27.Patient Health Risk Analysis

27.1. Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol. Examine pairwiserelationships to identify strong correlations, clusters, or potential risk patterns.

```
patient <- data.frame(  
  Patient_ID = c("P1","P2","P3","P4","P5"),  
  Age = c(25,40,55,35,60),  
  BMI = c(22,28,30,26,32),  
  BP = c(120,135,145,130,150),  
  Cholesterol = c(180,210,240,200,260)  
)  
  
pairs(patient[,2:5],  
  main = "Scatterplot Matrix of Health Indicators",  
  col = "blue",  
  pch = 19)
```



27.2. Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.

Q-Q Plot for Cholesterol

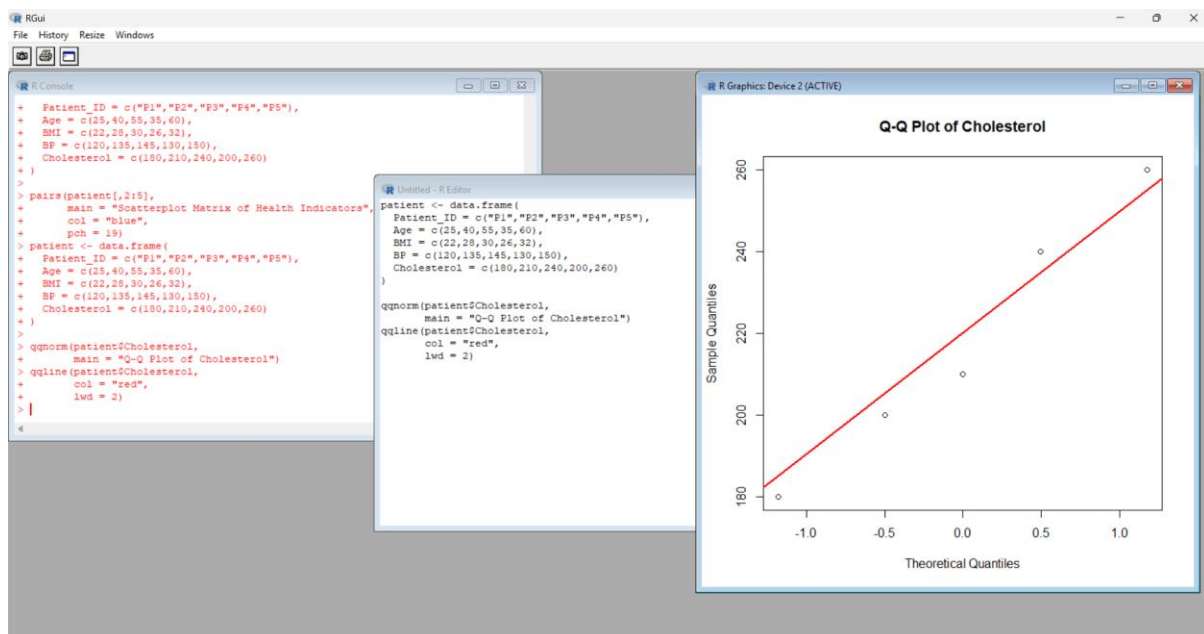
```
patient <- data.frame(  
  Patient_ID = c("P1", "P2", "P3", "P4", "P5"),  
  Age = c(25, 40, 55, 35, 60),  
  BMI = c(22, 28, 30, 26, 32),  
  BP = c(120, 135, 145, 130, 150),  
  Cholesterol = c(180, 210, 240, 200, 260)  
)
```

```
qqnorm(patient$Cholesterol,  
  main = "Q-Q Plot of Cholesterol")
```

```
qqline(patient$Cholesterol,
```

```
  col = "red",
```

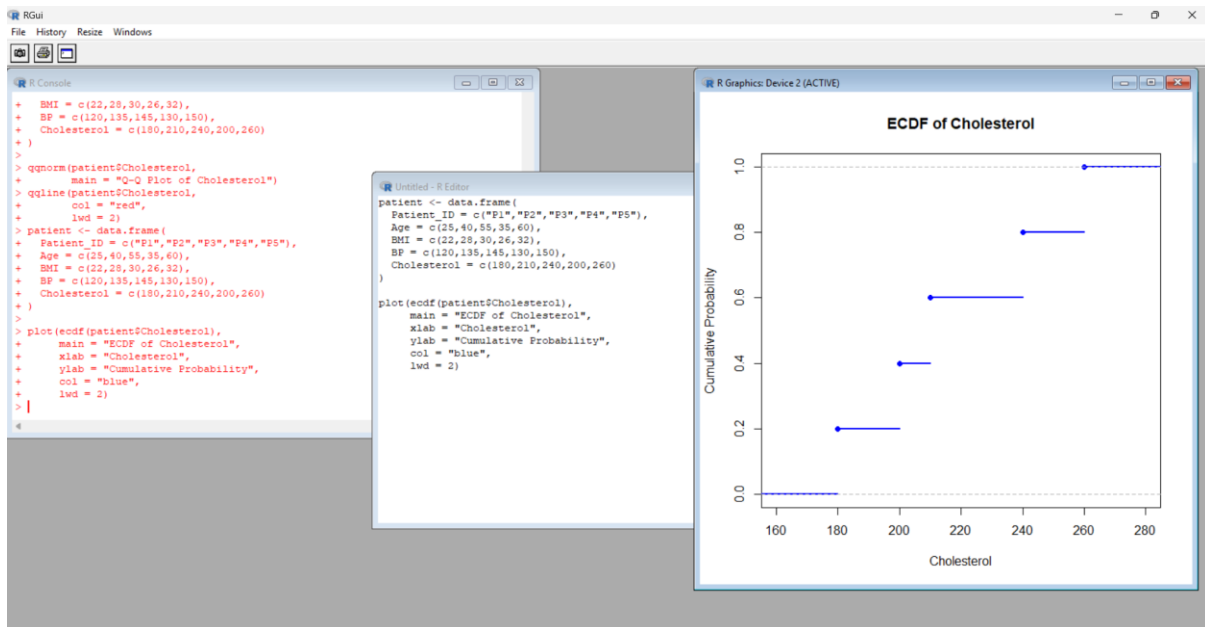
```
  lwd = 2)
```



ECDF Plot

```
patient <- data.frame(  
  Patient_ID = c("P1","P2","P3","P4","P5"),  
  Age = c(25,40,55,35,60),  
  BMI = c(22,28,30,26,32),  
  BP = c(120,135,145,130,150),  
  Cholesterol = c(180,210,240,200,260)  
)
```

```
plot(ecdf(patient$Cholesterol),  
     main = "ECDF of Cholesterol",  
     xlab = "Cholesterol",  
     ylab = "Cumulative Probability",  
     col = "blue",  
     lwd = 2)
```

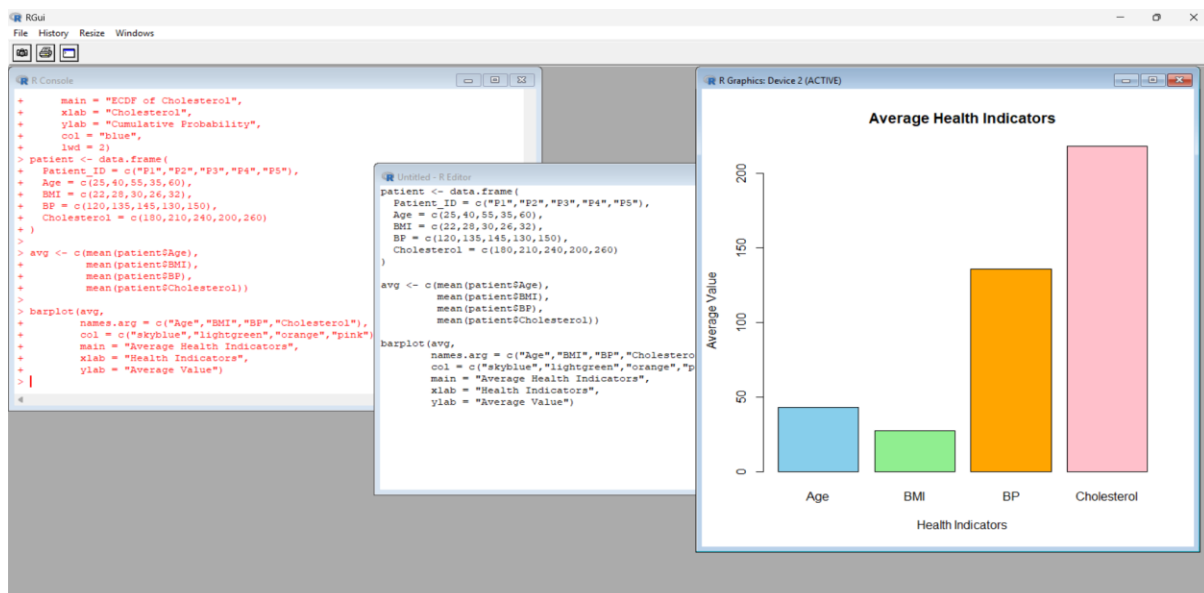


27.3. Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk

```
patient <- data.frame(
  Patient_ID = c("P1", "P2", "P3", "P4", "P5"),
  Age = c(25, 40, 55, 35, 60),
  BMI = c(22, 28, 30, 26, 32),
  BP = c(120, 135, 145, 130, 150),
  Cholesterol = c(180, 210, 240, 200, 260)
)

avg <- c(mean(patient$Age),
  mean(patient$BMI),
  mean(patient$BP),
  mean(patient$Cholesterol))

barplot(avg,
  names.arg = c("Age", "BMI", "BP", "Cholesterol"),
  col = c("skyblue", "lightgreen", "orange", "pink"),
  main = "Average Health Indicators",
  xlab = "Health Indicators",
  ylab = "Average Value")
```



27.4. Create an interactive Tableau dashboard with a heatmap highlighting high-risk patients and an Age filter for exploring risk patterns. Include hover tooltips showing all health indicators. Analyze which patients are highest risk, how age affects risk, and how this dashboard can aid healthcare professionals in prioritizing care.

```
patient <- data.frame(  
  Patient_ID = c("P1", "P2", "P3", "P4", "P5"),  
  Age = c(25, 40, 55, 35, 60),  
  BMI = c(22, 28, 30, 26, 32),  
  BP = c(120, 135, 145, 130, 150),  
  Cholesterol = c(180, 210, 240, 200, 260)  
)
```

```
par(mfrow = c(1, 2))
```

```
plot(patient$Age,  
  patient$Cholesterol,  
  pch = 19,  
  col = "blue",  
  main = "Age vs Cholesterol",  
  xlab = "Age",  
  ylab = "Cholesterol")
```

```
avg <- c(mean(patient$Age),  
  mean(patient$BMI),  
  mean(patient$BP),  
  mean(patient$Cholesterol))
```

```
barplot(avg,  
  names.arg = c("Age", "BMI", "BP", "Cholesterol"),
```

```

col = c("skyblue", "lightgreen", "orange", "pink"),

main = "Average Health Indicators",

ylab = "Average Value")

par(mfrow = c(1,1))

```



SET 28. Vehicle Performance Analysis

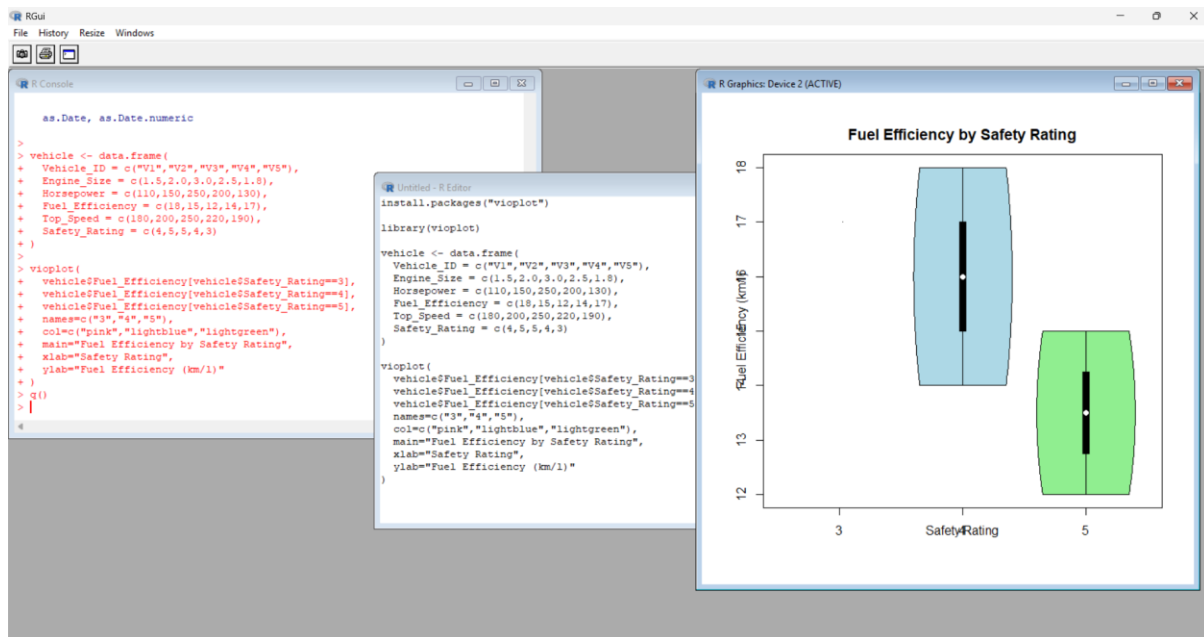
28.1. Using R, develop a violin plot to compare the distribution of Fuel Efficiency across different Safety Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

```
install.packages("vioplot")
```

```
library(vioplot)
```

```
vehicle <- data.frame(  
  Vehicle_ID = c("V1", "V2", "V3", "V4", "V5"),  
  Engine_Size = c(1.5, 2.0, 3.0, 2.5, 1.8),  
  Horsepower = c(110, 150, 250, 200, 130),  
  Fuel_Efficiency = c(18, 15, 12, 14, 17),  
  Top_Speed = c(180, 200, 250, 220, 190),  
  Safety_Rating = c(4, 5, 5, 4, 3)  
)
```

```
vioplot(  
  vehicle$Fuel_Efficiency[vehicle$Safety_Rating==3],  
  vehicle$Fuel_Efficiency[vehicle$Safety_Rating==4],  
  vehicle$Fuel_Efficiency[vehicle$Safety_Rating==5],  
  names=c("3", "4", "5"),  
  col=c("pink", "lightblue", "lightgreen"),  
  main="Fuel Efficiency by Safety Rating",  
  xlab="Safety Rating",  
  ylab="Fuel Efficiency (km/l)"  
)
```



28.2. Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine Size. Analyse the strength and direction of correlation and find observable performance trends.

```

vehicle <- data.frame(
  Vehicle_ID = c("V1", "V2", "V3", "V4", "V5"),
  Engine_Size = c(1.5, 2.0, 3.0, 2.5, 1.8),
  Horsepower = c(110, 150, 250, 200, 130),
  Fuel_Efficiency = c(18, 15, 12, 14, 17),
  Top_Speed = c(180, 200, 250, 220, 190),
  Safety_Rating = c(4, 5, 5, 4, 3)
)

```

```

color <- ifelse(vehicle$Engine_Size < 2, "blue",
  ifelse(vehicle$Engine_Size < 2.5, "green", "red"))

```

```

plot(vehicle$Horsepower,
  vehicle$Top_Speed,

```

```

pch=19,

col=color,

main="Horsepower vs Top Speed",

xlab="Horsepower",

ylab="Top Speed")

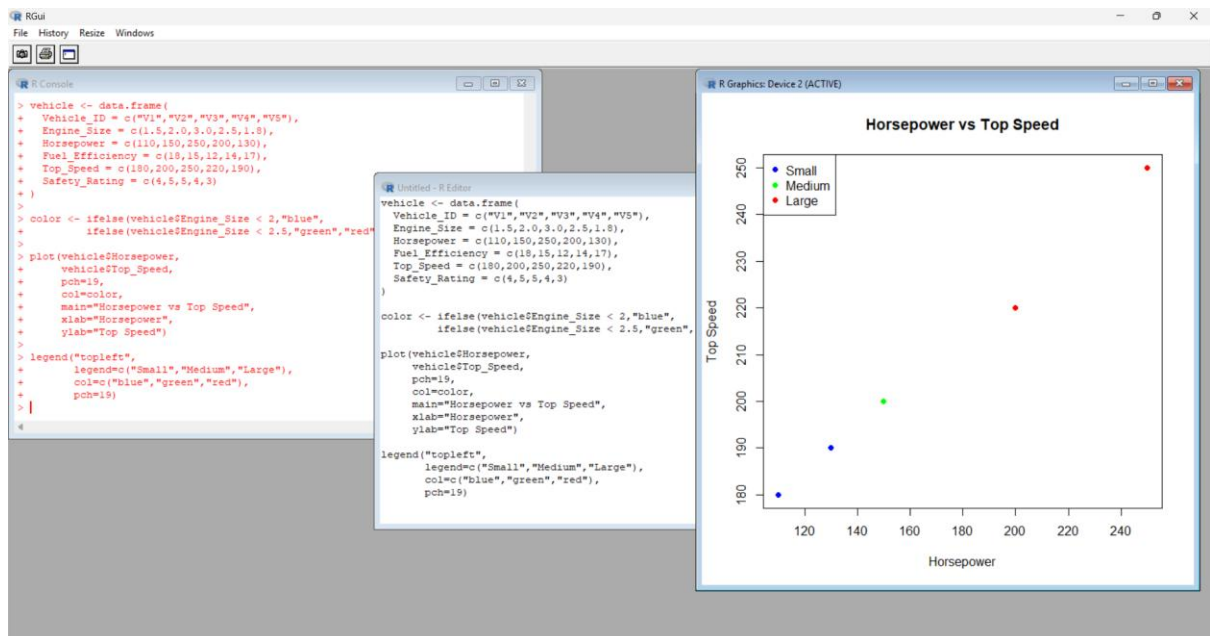
legend("topleft",

      legend=c("Small", "Medium", "Large"),

      col=c("blue", "green", "red"),

pch=19)

```

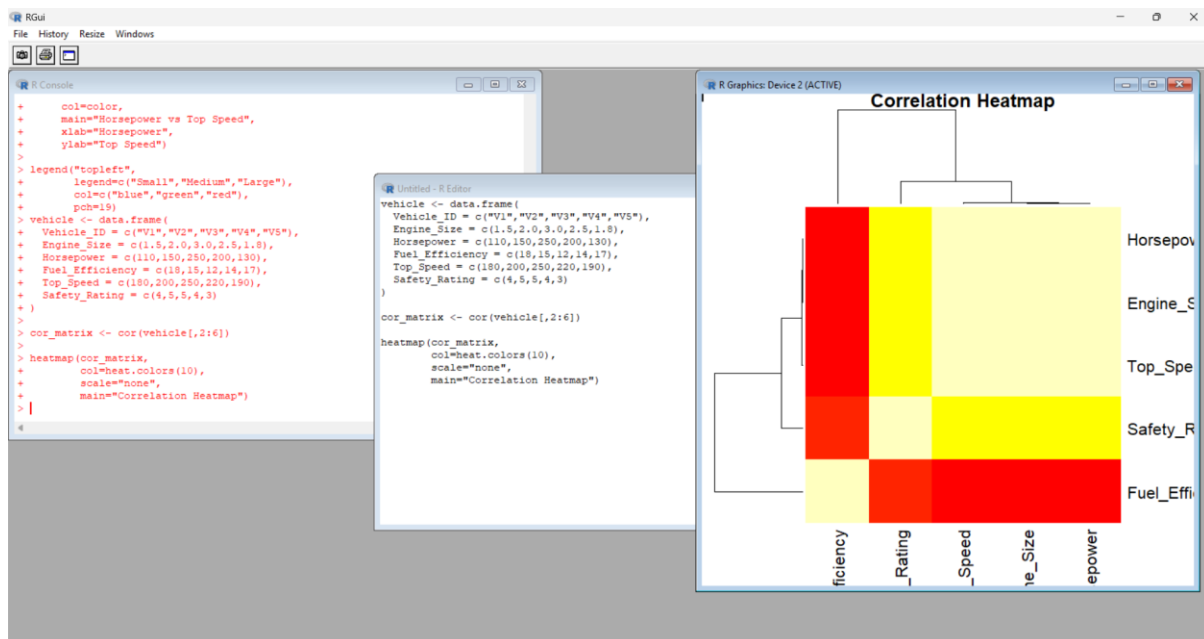


28.3. Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.

```
vehicle <- data.frame(
  Vehicle_ID = c("V1","V2","V3","V4","V5"),
  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),
  Horsepower = c(110,150,250,200,130),
  Fuel_Efficiency = c(18,15,12,14,17),
  Top_Speed = c(180,200,250,220,190),
  Safety_Rating = c(4,5,5,4,3)
)
```

```
cor_matrix<- cor(vehicle[,2:6])
```

```
heatmap(cor_matrix,
  col=heat.colors(10),
  scale="none",
  main="Correlation Heatmap")
```



28.4. In Tableau, design an interactive dashboard incorporating bar charts and scatter plots, along with filters for Engine Size and Safety Rating, to highlight vehicles that balance fuel efficiency and high performance. Provide analytical interpretation of the insights derived from the dashboard.

```
vehicle <- data.frame(  
  Vehicle_ID = c("V1", "V2", "V3", "V4", "V5"),  
  Engine_Size = c(1.5, 2.0, 3.0, 2.5, 1.8),  
  Horsepower = c(110, 150, 250, 200, 130),  
  Fuel_Efficiency = c(18, 15, 12, 14, 17),  
  Top_Speed = c(180, 200, 250, 220, 190),  
  Safety_Rating = c(4, 5, 5, 4, 3)  
)
```

```
par(mfrow=c(1,2))
```

```
barplot(vehicle$Fuel_Efficiency,  
  names.arg=vehicle$Vehicle_ID,  
  col="lightgreen",  
  main="Fuel Efficiency",  
  xlab="Vehicle",  
  ylab="km/l")
```

```
plot(vehicle$Horsepower,  
  vehicle$Top_Speed,  
  pch=19,  
  col="blue",  
  main="Horsepower vs Top Speed",  
  xlab="Horsepower",  
  ylab="Top Speed")
```

```
par(mfrow=c(1,1))
```



SET 29 .Student Academic Performance

29.1. Using R, create a stacked area chart showing Test Score and Participation Score across students. Analyze trends.

```
student <- data.frame(
  Student_ID = c("S1","S2","S3","S4","S5"),
  Age = c(19,21,20,22,23),
  Study_Hours = c(12,8,15,10,7),
  Attendance = c(90,70,95,85,60),
  Test_Score = c(85,70,92,80,65),
  Participation_Score = c(8,7,9,8,6)
)
```

```
x <- 1:5
```

```
y1 <- student$Test_Score
```

```
y2 <- student$Participation_Score
```

```

plot(x, y1+y2,
     type="n",
     xlab="Students",
     ylab="Score",
     main="Stacked Area Chart")

```

```

polygon(c(x,rev(x)),
        c(rep(0,5),rev(y1)),
        col="skyblue")

```

```

polygon(c(x,rev(x)),
        c(y1,rev(y1+y2)),
        col="pink")

```

```

axis(1,at=x,labels=student$Student_ID)

```

```

legend("topleft",
       legend=c("Test Score","Participation Score"),
       fill=c("skyblue","pink"))

```

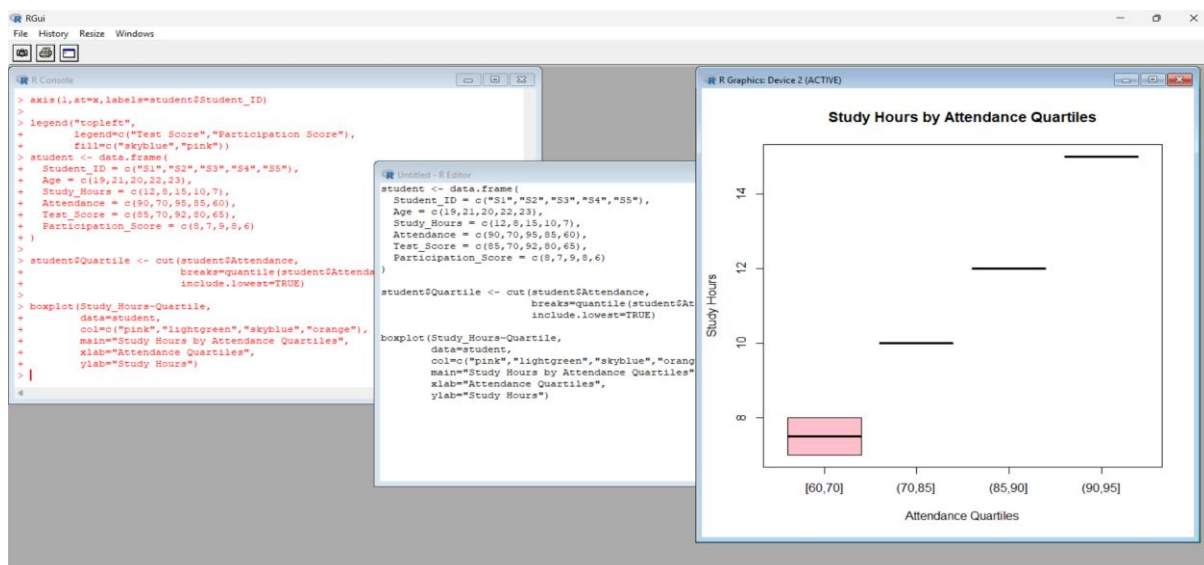


29.2. Using R, generate a boxplot of Study Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.

```
student <- data.frame(
  Student_ID = c("S1", "S2", "S3", "S4", "S5"),
  Age = c(19, 21, 20, 22, 23),
  Study_Hours = c(12, 8, 15, 10, 7),
  Attendance = c(90, 70, 95, 85, 60),
  Test_Score = c(85, 70, 92, 80, 65),
  Participation_Score = c(8, 7, 9, 8, 6)
)

student$Quartile <- cut(student$Attendance,
  breaks = quantile(student$Attendance),
  include.lowest = TRUE)

boxplot(Study_Hours ~ Quartile,
  data = student,
  col = c("pink", "lightgreen", "skyblue", "orange"),
  main = "Study Hours by Attendance Quartiles",
  xlab = "Attendance Quartiles",
  ylab = "Study Hours")
```



29.3. Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

```
student <- data.frame(  
  Student_ID = c("S1", "S2", "S3", "S4", "S5"),  
  Age = c(19, 21, 20, 22, 23),  
  Study_Hours = c(12, 8, 15, 10, 7),  
  Attendance = c(90, 70, 95, 85, 60),  
  Test_Score = c(85, 70, 92, 80, 65),  
  Participation_Score = c(8, 7, 9, 8, 6)  
)  
d <- density(student$Test_Score)  
plot(d,  
  col="blue",  
  lwd=2,  
  main="Density Plot of Test Scores",  
  xlab="Test Score",  
  ylab="Density")  
polygon(d,  
  col="lightblue")
```



29.4. In Tableau, design an interactive scatterplot with Study Hours on X-axis, Test Score on Y-axis, colored by Participation Score. Include filters for Age.

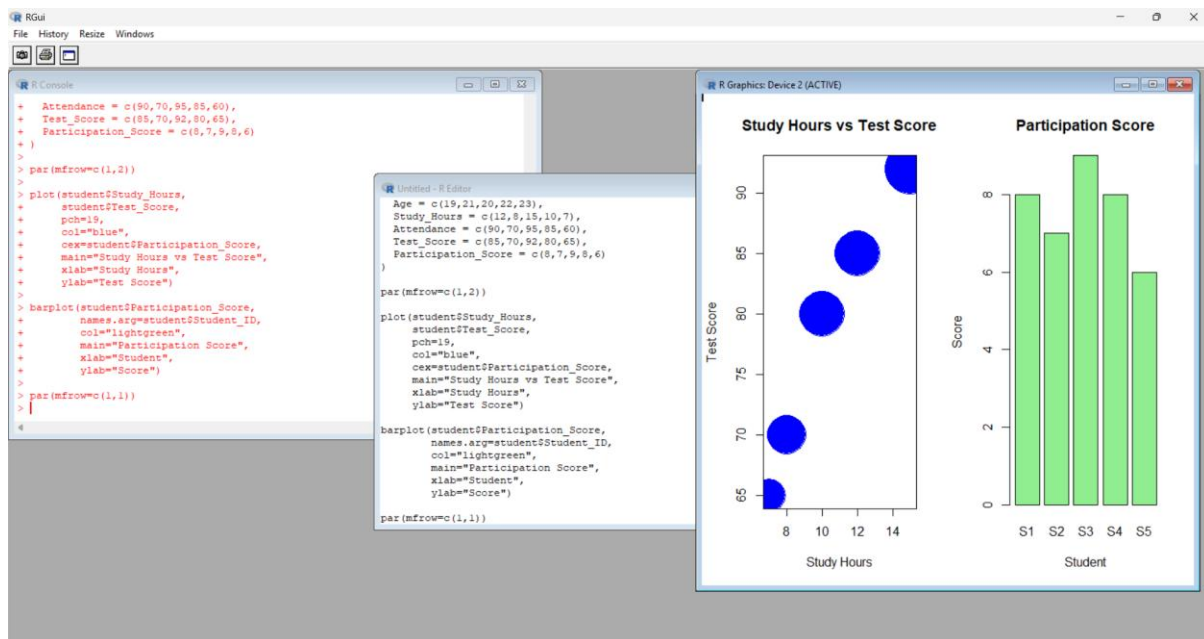
```
student <- data.frame(  
  Student_ID = c("S1", "S2", "S3", "S4", "S5"),  
  Age = c(19, 21, 20, 22, 23),  
  Study_Hours = c(12, 8, 15, 10, 7),  
  Attendance = c(90, 70, 95, 85, 60),  
  Test_Score = c(85, 70, 92, 80, 65),  
  Participation_Score = c(8, 7, 9, 8, 6)  
)
```

```
par(mfrow=c(1,2))
```

```
plot(student$Study_Hours,  
  student$Test_Score,  
  pch=19,  
  col="blue",  
  cex=student$Participation_Score,  
  main="Study Hours vs Test Score",  
  xlab="Study Hours",  
  ylab="Test Score")
```

```
barplot(student$Participation_Score,  
  names.arg=student$Student_ID,  
  col="lightgreen",  
  main="Participation Score",  
  xlab="Student",  
  ylab="Score")
```

```
par(mfrow=c(1,1))
```



SET 30

30.1. Import the dataset in R. Plot a histogram and density plot for Screen Time. Add suitable colors and labels. Interpret the screen-time distribution.

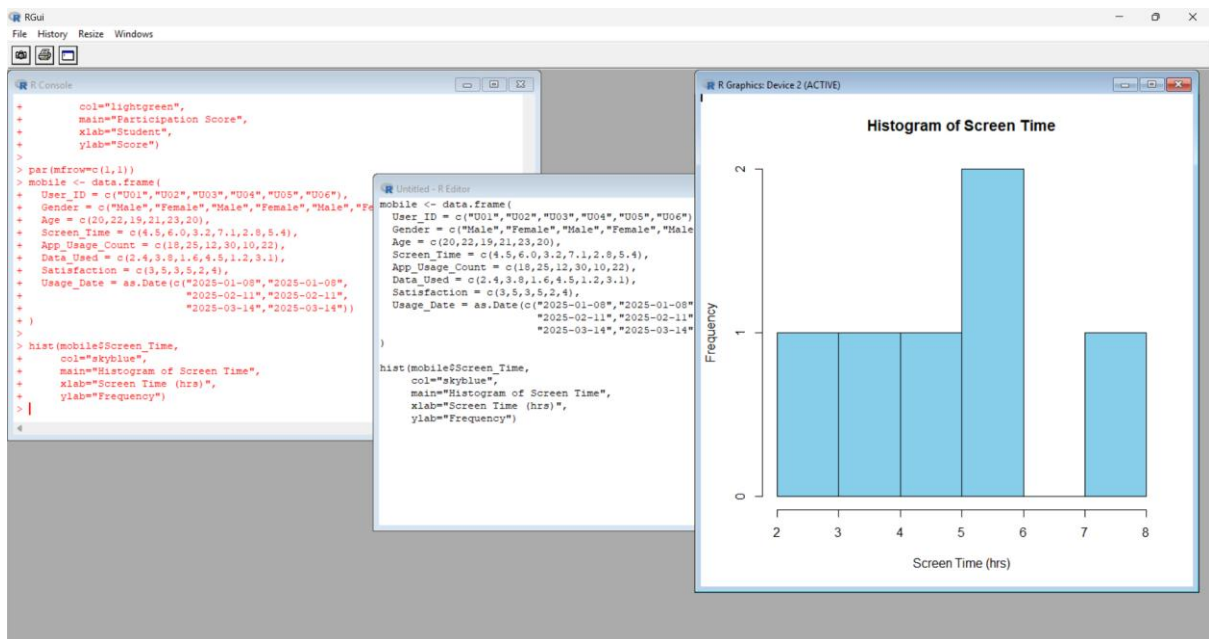
Histogram for Screen Time

```
mobile <- data.frame(
  User_ID = c("U01","U02","U03","U04","U05","U06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),
  App_Usage_Count = c(18,25,12,30,10,22),
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),
  Satisfaction = c(3,5,3,5,2,4),
  Usage_Date = as.Date(c("2025-01-08","2025-01-08",
    "2025-02-11","2025-02-11",
```


"2025-03-14","2025-03-14"))

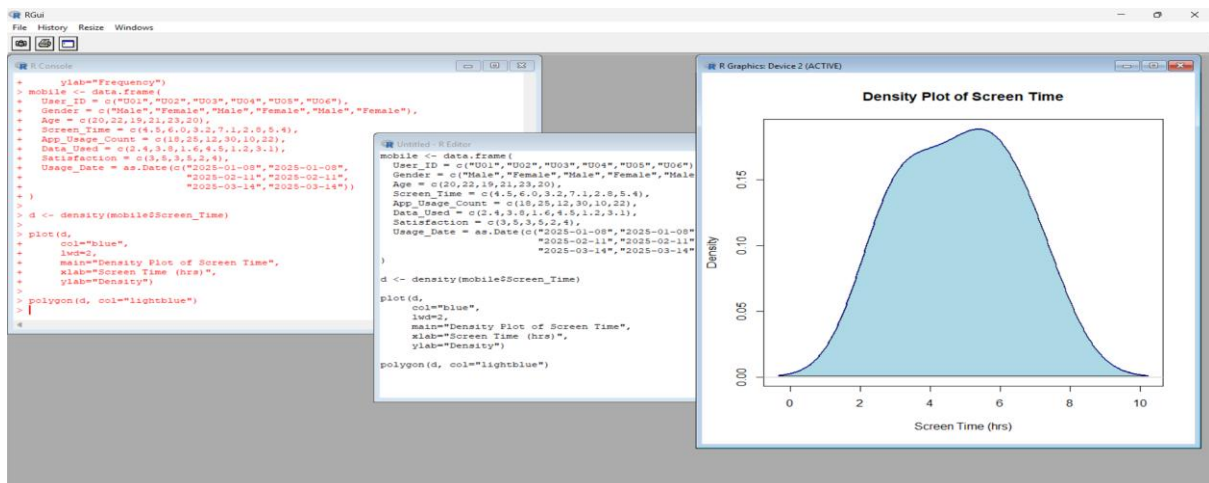
)

```
hist(mobile$Screen_Time,  
     col="skyblue",  
     main="Histogram of Screen Time",  
     xlab="Screen Time (hrs)",  
     ylab="Frequency")
```



Density Plot for Screen Time

```
mobile <- data.frame(  
  User_ID = c("U01","U02","U03","U04","U05","U06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Age = c(20,22,19,21,23,20),  
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),  
  App_Usage_Count = c(18,25,12,30,10,22),  
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),  
  Satisfaction = c(3,5,3,5,2,4),  
  Usage_Date = as.Date(c("2025-01-08","2025-01-08",  
    "2025-02-11","2025-02-11",  
    "2025-03-14","2025-03-14"))  
)  
d <- density(mobile$Screen_Time)  
plot(d,  
  col="blue",  
  lwd=2,  
  main="Density Plot of Screen Time",  
  xlab="Screen Time (hrs)",  
  ylab="Density")  
polygon(d, col="lightblue")
```



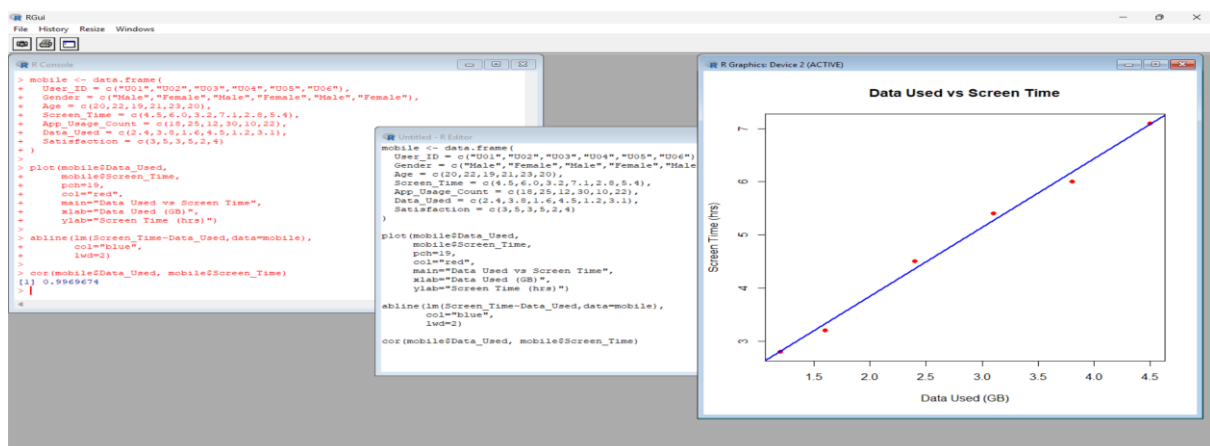
30.2. Create a scatter plot using R programming between Data Used and Screen Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

```
mobile <- data.frame(
  User_ID = c("U01","U02","U03","U04","U05","U06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),
  App_Usage_Count = c(18,25,12,30,10,22),
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),
  Satisfaction = c(3,5,3,5,2,4)
)

plot(mobile$Data_Used,
     mobile$Screen_Time,
     pch=19,
     col="red",
     main="Data Used vs Screen Time",
     xlab="Data Used (GB)",
     ylab="Screen Time (hrs)")

abline(lm(Screen_Time~Data_Used,data=mobile),
       col="blue",
       lwd=2)

cor(mobile$Data_Used, mobile$Screen_Time)
```



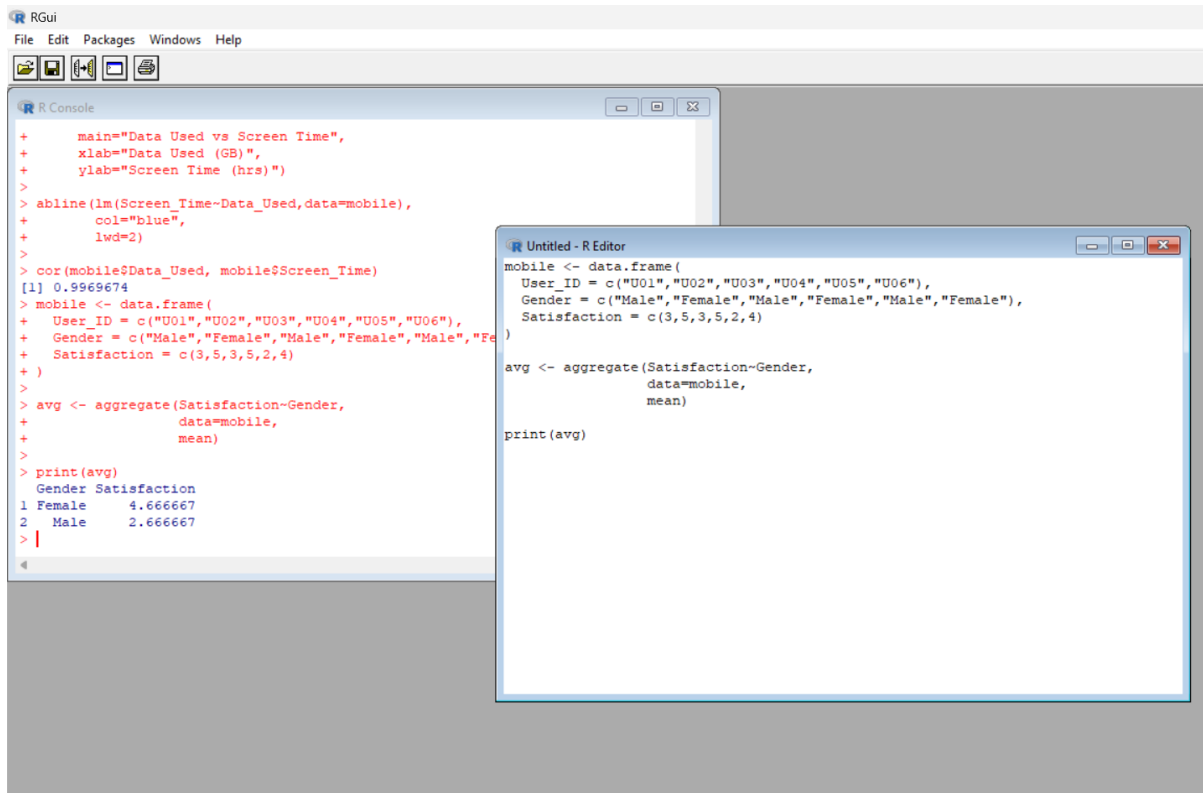
30.3. Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

Average Satisfaction by Gender

```
mobile <- data.frame(  
  User_ID = c("U01","U02","U03","U04","U05","U06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Satisfaction = c(3,5,3,5,2,4)  
)
```

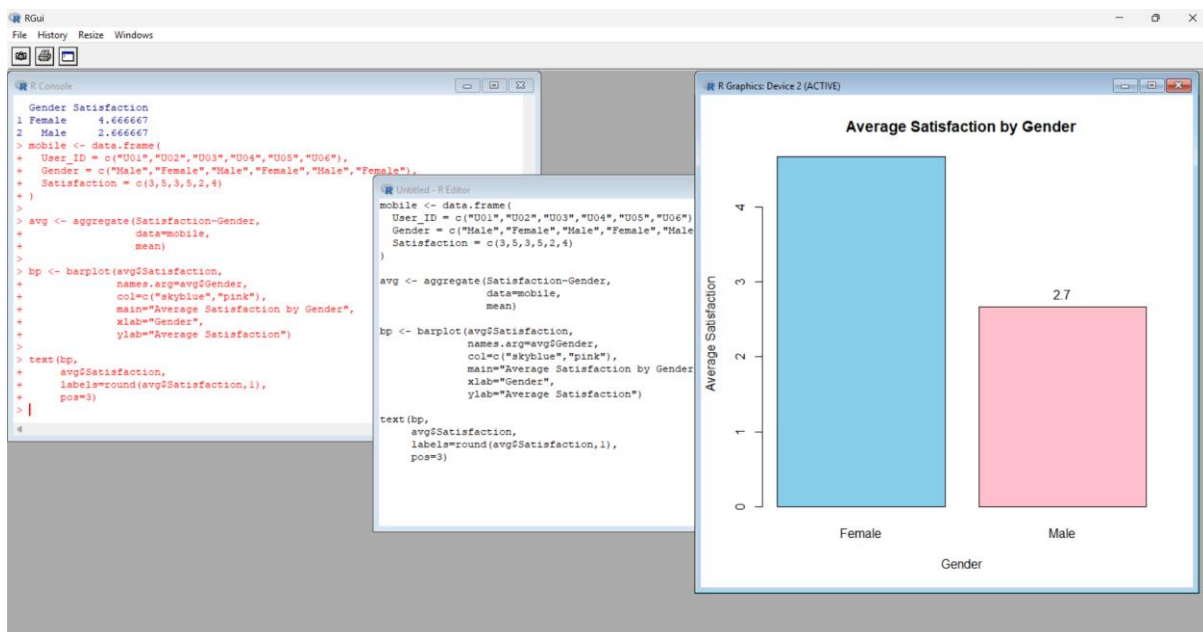
```
avg<- aggregate(Satisfaction~Gender,  
  data=mobile,  
  mean)
```

```
print(avg)
```



Bar Chart for Average Satisfaction

```
mobile <- data.frame(  
  User_ID = c("U01","U02","U03","U04","U05","U06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Satisfaction = c(3,5,3,5,2,4)  
)  
  
avg<- aggregate(Satisfaction~Gender,  
  data=mobile,  
  mean)  
  
bp <- barplot(avg$Satisfaction,  
  names.arg=avg$Gender,  
  col=c("skyblue","pink"),  
  main="Average Satisfaction by Gender",  
  xlab="Gender",  
  ylab="Average Satisfaction")  
  
text(bp,  
  avg$Satisfaction,  
  labels=round(avg$Satisfaction,1),  
  pos=3)
```



30.4. Import Mobile_App_Usage.csv into Tableau. Create a bubble chart (Screen Time vs Data Used). Use color for Gender and size for App Usage Count. Create a line chart for usage trends. Add filters for Gender and Usage Date. Design a mini dashboard.

```
mobile <- data.frame(  
  User_ID = c("U01","U02","U03","U04","U05","U06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),  
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),  
  App_Usage_Count = c(18,25,12,30,10,22)  
)
```

```
par(mfrow=c(1,2))
```

```
plot(mobile$Screen_Time,  
     mobile$Data_Used,  
     pch=19,  
     cex=mobile$App_Usage_Count/10,  
     col="blue",  
     main="Screen Time vs Data Used",  
     xlab="Screen Time (hrs)",  
     ylab="Data Used (GB)")
```

```
barplot(mobile$App_Usage_Count,  
        names.arg=mobile$User_ID,  
        col="lightgreen",  
        main="App Usage Count",  
        xlab="User",  
        ylab="Usage Count")
```

```
par(mfrow=c(1,1))
```

